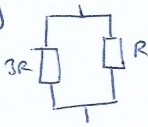


Etude d'un circuit électrique.

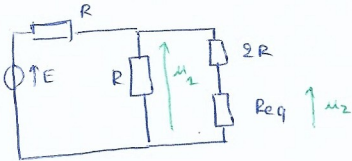
1)



$$\frac{1}{R_{eq}} = \frac{1}{3R} + \frac{1}{R} = \frac{1}{3R} + \frac{3}{3R} = \frac{4}{3R}$$

$$\Rightarrow R_{eq} = \frac{3}{4} R$$

2) On a 1 p^e div de tension :



$$u_2 = \frac{R_{eq}}{R_{eq} + 2R} u_1$$

$$= \frac{3/4 R}{3/4 R + 2R} u_1$$

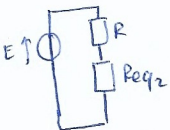
$$= \frac{3/4}{3/4 + 2} u_1$$

$$= \frac{3}{3 + 8} u_1$$

$$= \frac{3}{11} u_1$$

$$u_2 = \frac{3}{11} u_1$$

3)

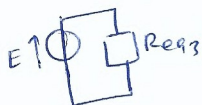


$$R_{eq2} = (2R + R_{eq}) \parallel R$$

$$\frac{1}{R_{eq2}} = \frac{1}{2R + 3/4 R} + \frac{1}{R} = \frac{1}{11/4 R} + \frac{1}{R} = \frac{4}{11R} + \frac{11}{11R}$$

$$= \frac{15}{11R}$$

$$R_{eq2} = \frac{11R}{15}$$



$$R_{eq3} = \frac{11R}{15} + R \times \frac{15}{15} = \frac{26R}{15}$$

$$\Rightarrow E = R_{eq3} I$$

$$\Rightarrow I = \frac{E}{R_{eq3}} = \frac{15}{26} \frac{E}{R}$$

On en déduit

$$\bullet U_1 = E - RI$$

$$= E - \frac{15}{26} \frac{E}{R} \times R$$

$$U_1 = \frac{11}{26} E$$

$$\Rightarrow I_1 = \frac{U_1}{R} = \frac{11}{26} \frac{E}{R}$$

$$\bullet U_2 = \frac{3}{11} U_1 = \frac{3}{11} \cdot \frac{11}{26} E$$

$$U_2 = \frac{3}{26} E$$

$$\Rightarrow I_4 = \frac{U_2}{R} = \frac{3}{26} \frac{E}{R}$$

$$\Rightarrow I_3 = \frac{U_2}{3R} = \frac{3}{26} \frac{E}{3R} = \frac{E}{26R}$$

$$\bullet I_2 = I_3 + I_4$$

$$= \frac{E}{26R} + \frac{3E}{26R} = \frac{E}{R} \times \frac{2}{13} =$$

$$\frac{2}{13} \frac{E}{R} = I_2$$

$$4) \quad P_{\text{gén}} = EI$$

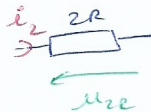
$$P_{\text{gén}} = E \times \frac{15}{26} \frac{E}{R}$$

$$P_{\text{gén}} = \frac{E^2}{R} \cdot \frac{15}{26}$$

$P_{\text{gén}} > 0$ car il fournit de l'énergie élec. au circuit.

$$\begin{aligned} \text{S)} \quad P_{2R} &= 2R \cdot (i_2)^2 \\ &= 2R \left(\frac{2}{13} \frac{E}{R} \right)^2 \end{aligned}$$

$$P_{2R} = \frac{8}{169} \frac{E^2}{R}$$



P_{2R} est positive & la résistance reçoit de l'énergie élec (c'est-à-dire qu'elle agit comme un récepteur passif).